

Supplementary Material

Supplementary Table 1. Sex and age of all PBMC controls.

		Cutaneous melanoma	Uveal melanoma	Healthy donors
Sex	Females	3 (21%)	7 (50%)	8 (62%)
	Males	11 (79%)	7 (50%)	5 (38%)
	Total	14	14	13
Age	(Median + range)			
	Females	73 years (71-79)	71 years (45-74)	74 years (71-75)
	Males	75 years (56-87)	65 years (55-80)	66 years (65-78)
	Total	75 years	68 years	73 years

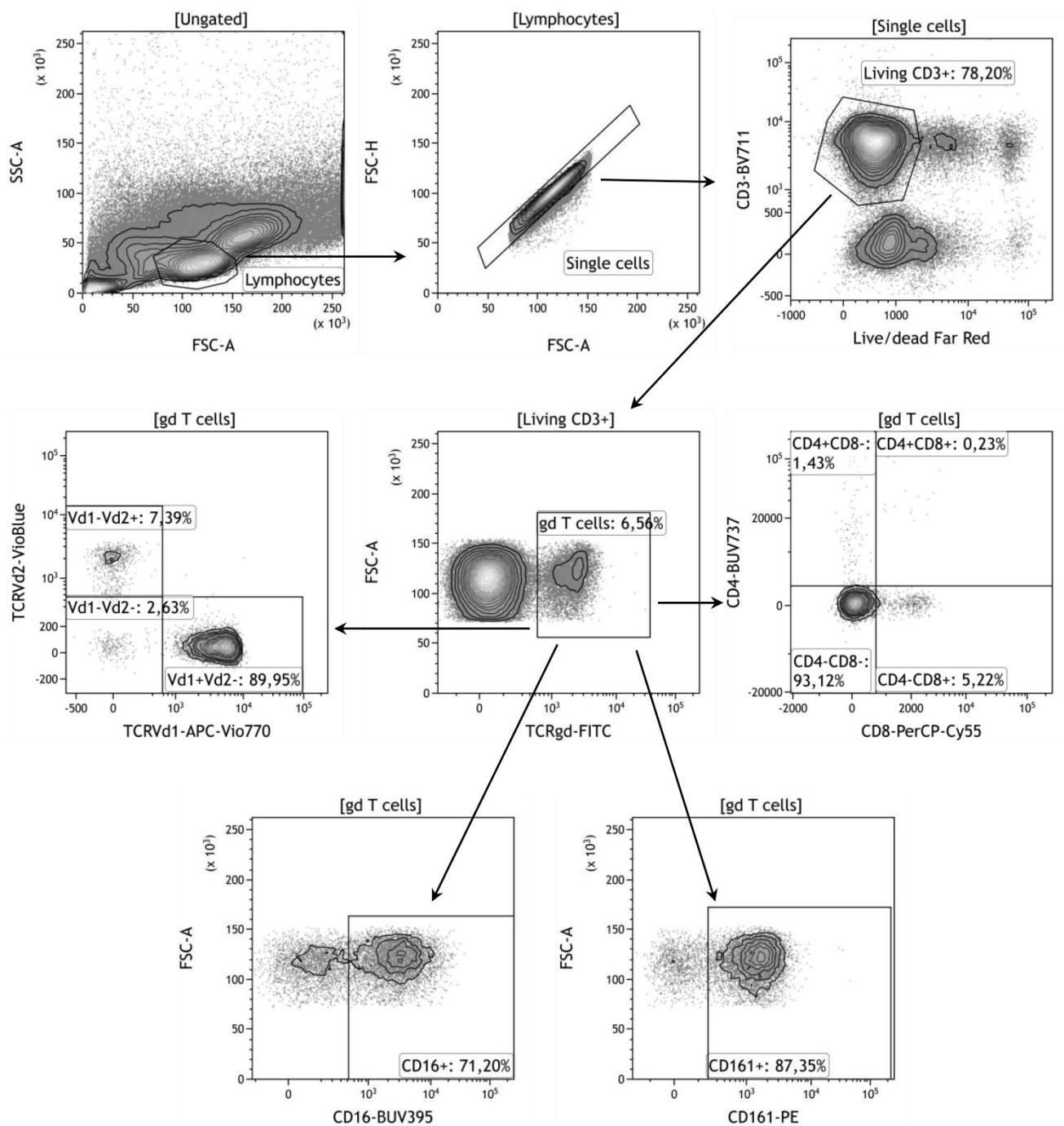
Supplementary Table 2. Comparison of nucleotide sequences of CALPI-like CDR3 clonotypes and *TRDJ* primers conclude the former are an artifact and should not be included in the analysis.

Name	Sequence (nt.)
CALPI-like clonotype	TGTGCTCTTCCGATCT
TRDJ1	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTCACAGTCACACGGG TTCCTT
TRDJ2	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTCGATGAGTTGTGTT CCCTTTCCAA
TRDJ3	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTAGTTTGATGCCAGT TCCGAAA
TRDJ4	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTGTTGTACCTCCAGA TAGGTTTCCT

Supplementary Table 3. List of conjugated antibodies for flow cytometry.

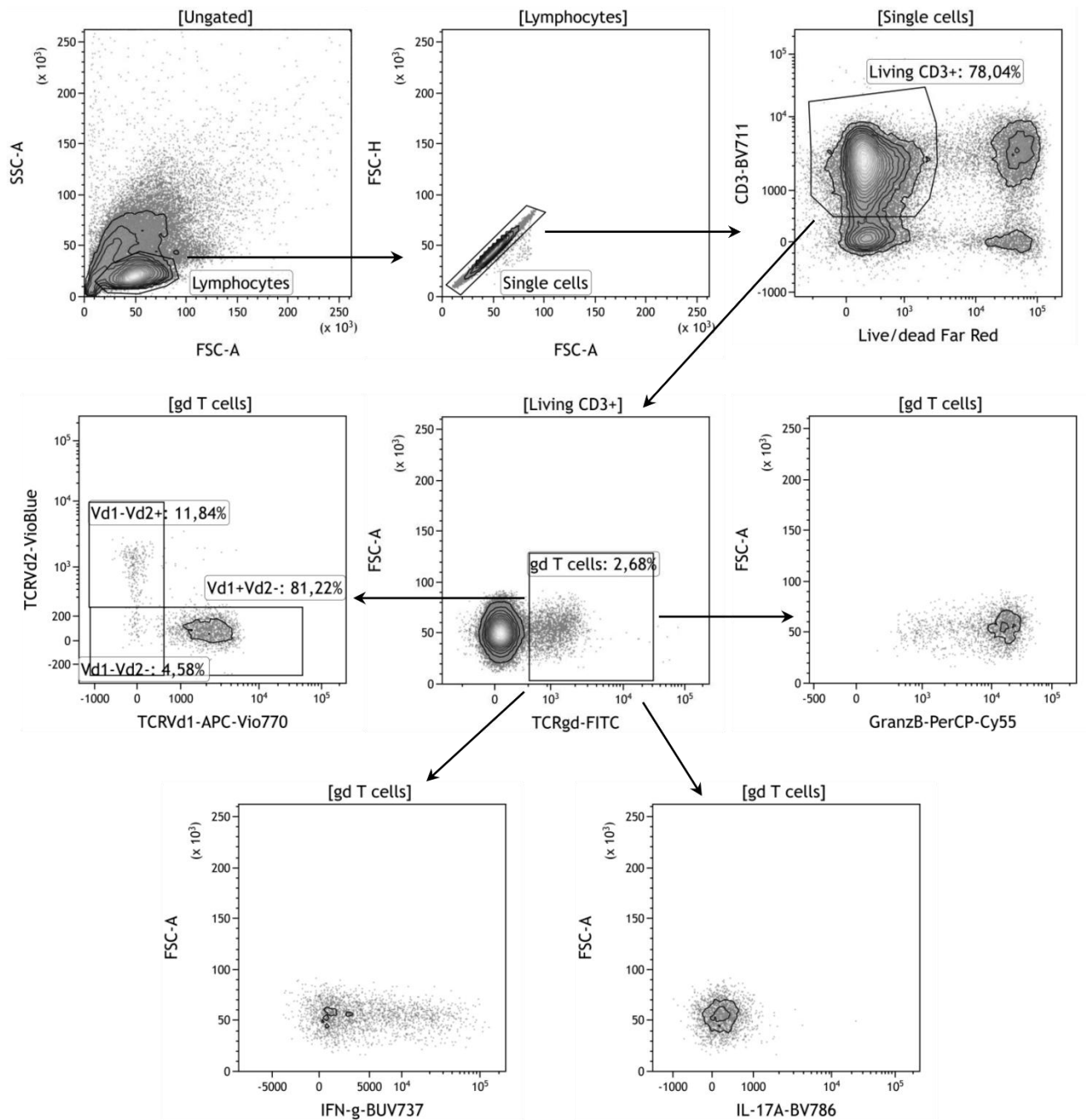
Panel*	Antigen	Fluorophore	Clone	Manufacturer
Unstim, Stim	CD3	BV711	UCHT1	BD Biosciences (#563725)
Unstim	CD4	BUV737	OKT-4	BD Biosciences (#750977)
Unstim	CD8	PerCP-Cy5.5	RPA-T8	BD Biosciences (#560662)
Unstim, Stim	CD16	BUV395	3G8	BD Biosciences (#563785)
Unstim, Stim	CD161	PE	HP-3G10	BD Biosciences (#566843)
Unstim, Stim	NKG2D	PE-Cy7	1D11	BD Biosciences (#562365)
Unstim, Stim	TCR γ/δ	FITC	11F2	Miltenyi Biotec (#130-114-029)
Unstim, Stim	TCRV δ 1	APC-Vio770	REA173	Miltenyi Biotec (#130-120-438)
Unstim, Stim	TCRV δ 2	VioBlue	123R3	Miltenyi Biotec (#130-101-152)
Stim	Granzyme B	PerCP-Cy5.5	QA16A02	BioLegend (#372211)
Stim	IFN- γ	BUV737	4S.B3	BD Biosciences (#612845)
Stim	IL-17A	BV786	N49-653	BD Biosciences (#563745)

*Unstim = unstimulated panel, Stim = Stimulated panel



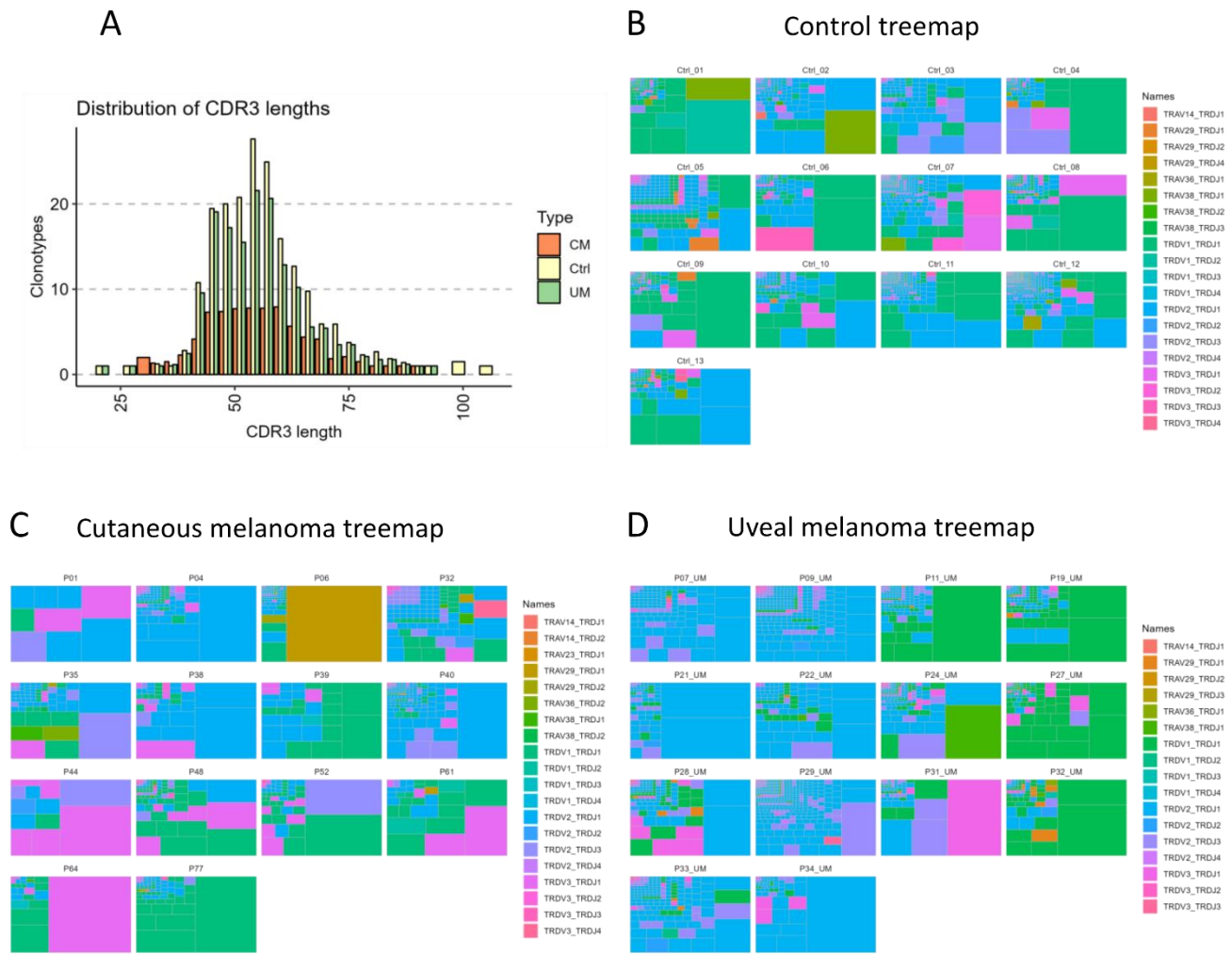
Supplementary Figure 1. Flow cytometry cell gating strategy for unstimulated samples

Flow cytometry plots showing the gating strategy for $\gamma\delta$ T cells in peripheral blood from a representative unstimulated sample.



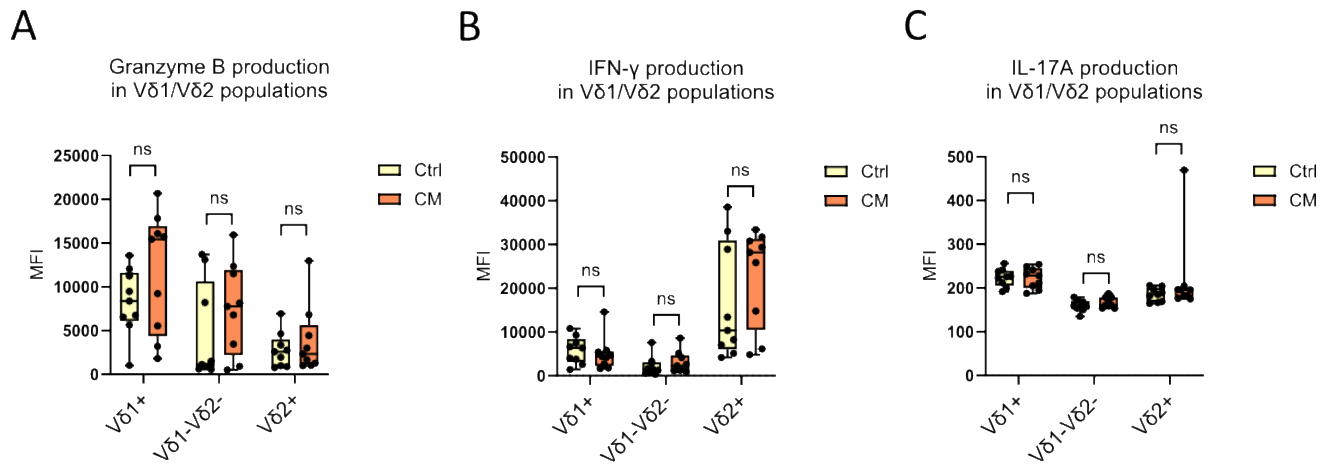
Supplementary Figure 2. Flow cytometry cell gating strategy for stimulated samples

Flow cytometry plots showing the gating strategy for $\gamma\delta$ T cells in peripheral blood from a representative stimulated sample.



Supplementary Figure 3. Group clonality distribution

(A) Distribution of clonotypes by CDR3 nucleotide lengths within each group, following an approximate Gaussian curve, which means that not an individual clonotype stands out per group. And treemaps of clonotypes per sample within the (B) control, (C) cutaneous melanoma, and (D) uveal melanoma groups. Colors show the combination of *TRDV* and *TRDJ* genes per clonotype. Ctrl=Control, CM=Cutaneous melanoma, UM=Uveal melanoma.



Supplementary Figure 4. Extended functionality studies

The intracellular expression of **(A)** granzyme B, **(B)** IFN-γ and **(C)** IL-17A, as measured by flow cytometry, in peripheral γδ T cells from cutaneous melanoma patients and healthy controls after 4 hour stimulation with PMA and ionomycin. $n_{CM}=9$, $n_{Ctrl}=9$, unpaired Mann-Whitney test, ns=not significant. MFI=Median fluorescence intensity, Ctrl=Control, CM=Cutaneous melanoma.